

# A Closed-Form Surjection from $\mathbb{N}$ onto $\mathbb{Q}$ — With Its Pseudo-Inverse

*Cantor Said the Rationals Are Countable. Here's an Actual Formula.*

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*Attribution note.* Sections 1–2, including the formulas (1)–(4) and the conjecture that  $r(s(p/q)) = p/q$ , are due to the author. The proofs in Sections 3–4 (Lemma 1 and Theorems 1–2) were found by Claude Fable 5, a large language model developed by Anthropic, acting as an automated theorem prover [3]; the author has verified the argument.

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## 1 Introduction

Throughout this document,  $\mathbb{N}$  denotes the set of positive integers  $\{1, 2, 3, \dots\}$ . If you were/are like I was when I first learned that  $\mathbb{N}$  is equipollent to  $\mathbb{Q}$ , you saw an argument in which the positive rationals are counted. In this document, I will reveal a formula that is a surjection of  $\mathbb{N}$  onto  $\mathbb{Q}$ . The first 100 outputs of this formula (at  $n = 1, 2, \dots, 100$ ) are as follows:

0, 1, -1,  $\frac{1}{2}$ ,  $-\frac{1}{2}$ , 2, -2,  $\frac{1}{3}$ ,  $-\frac{1}{3}$ , 1, -1, 3, -3,  $\frac{1}{4}$ ,  $-\frac{1}{4}$ ,  $\frac{2}{3}$ ,  $-\frac{2}{3}$ ,  $\frac{3}{2}$ ,  $-\frac{3}{2}$ , 4, -4,  $\frac{1}{5}$ ,  $-\frac{1}{5}$ ,  $\frac{1}{2}$ ,  $-\frac{1}{2}$ , 1, -1, 2, -2, 5, -5,  $\frac{1}{6}$ ,  $-\frac{1}{6}$ ,  $\frac{2}{5}$ ,  $-\frac{2}{5}$ ,  $\frac{3}{4}$ ,  $-\frac{3}{4}$ ,  $\frac{4}{3}$ ,  $-\frac{4}{3}$ ,  $\frac{5}{2}$ ,  $-\frac{5}{2}$ , 6, -6,  $\frac{1}{7}$ ,  $-\frac{1}{7}$ ,  $\frac{1}{3}$ ,  $-\frac{1}{3}$ ,  $\frac{3}{5}$ ,  $-\frac{3}{5}$ , 1, -1,  $\frac{5}{3}$ ,  $-\frac{5}{3}$ , 3, -3, 7, -7,  $\frac{1}{8}$ ,  $-\frac{1}{8}$ ,  $\frac{2}{7}$ ,  $-\frac{2}{7}$ ,  $\frac{1}{2}$ ,  $-\frac{1}{2}$ ,  $\frac{4}{5}$ ,  $-\frac{4}{5}$ ,  $\frac{5}{4}$ ,  $-\frac{5}{4}$ , 2, -2,  $\frac{7}{5}$ ,  $-\frac{7}{5}$ , 8, -8,  $\frac{1}{9}$ ,  $-\frac{1}{9}$ ,  $\frac{1}{4}$ ,  $-\frac{1}{4}$ ,  $\frac{3}{7}$ ,  $-\frac{3}{7}$ ,  $\frac{2}{3}$ ,  $-\frac{2}{3}$ , 1, -1,  $\frac{3}{2}$ ,  $-\frac{3}{2}$ ,  $\frac{7}{3}$ ,  $-\frac{7}{3}$ , 4, -4, 9, -9,  $\frac{1}{10}$ ,  $-\frac{1}{10}$ ,  $\frac{2}{9}$ ,  $-\frac{2}{9}$ ,  $\frac{3}{8}$ ,  $-\frac{3}{8}$ ,  $\frac{4}{7}$ ,  $-\frac{4}{7}$ ,  $\frac{5}{6}$

The same values are displayed below in tabular form, indexed by the input  $n$ :

| $n$ | $r(n)$ | $n$ | $r(n)$ | $n$ | $r(n)$        | $n$ | $r(n)$         |
|-----|--------|-----|--------|-----|---------------|-----|----------------|
| 1   | 0      | 26  | 1      | 51  | -1            | 76  | $\frac{1}{4}$  |
| 2   | 1      | 27  | -1     | 52  | $\frac{5}{3}$ | 77  | $-\frac{1}{4}$ |

| $n$ | $r(n)$         | $n$ | $r(n)$         | $n$ | $r(n)$         | $n$ | $r(n)$          |
|-----|----------------|-----|----------------|-----|----------------|-----|-----------------|
| 3   | -1             | 28  | 2              | 53  | $-\frac{5}{3}$ | 78  | $\frac{3}{7}$   |
| 4   | $\frac{1}{2}$  | 29  | -2             | 54  | 3              | 79  | $-\frac{1}{3}$  |
| 5   | $-\frac{1}{2}$ | 30  | 5              | 55  | -3             | 80  | $\frac{2}{3}$   |
| 6   | 2              | 31  | -5             | 56  | 7              | 81  | $-\frac{2}{3}$  |
| 7   | -2             | 32  | $\frac{1}{6}$  | 57  | -7             | 82  | 1               |
| 8   | $\frac{1}{3}$  | 33  | $-\frac{1}{6}$ | 58  | $\frac{1}{8}$  | 83  | -1              |
| 9   | $-\frac{1}{3}$ | 34  | $\frac{2}{5}$  | 59  | $-\frac{1}{8}$ | 84  | $\frac{3}{2}$   |
| 10  | 1              | 35  | $-\frac{2}{5}$ | 60  | $\frac{2}{7}$  | 85  | $-\frac{3}{2}$  |
| 11  | -1             | 36  | $\frac{3}{4}$  | 61  | $-\frac{2}{7}$ | 86  | $\frac{7}{3}$   |
| 12  | 3              | 37  | $-\frac{3}{4}$ | 62  | $\frac{1}{2}$  | 87  | $-\frac{7}{3}$  |
| 13  | -3             | 38  | $\frac{4}{3}$  | 63  | $-\frac{1}{2}$ | 88  | 4               |
| 14  | $\frac{1}{4}$  | 39  | $-\frac{4}{3}$ | 64  | $\frac{4}{5}$  | 89  | -4              |
| 15  | $-\frac{1}{4}$ | 40  | $\frac{5}{2}$  | 65  | $-\frac{4}{5}$ | 90  | 9               |
| 16  | $\frac{2}{3}$  | 41  | $-\frac{5}{2}$ | 66  | $\frac{5}{4}$  | 91  | -9              |
| 17  | $-\frac{2}{3}$ | 42  | 6              | 67  | $-\frac{5}{4}$ | 92  | $\frac{1}{10}$  |
| 18  | $\frac{3}{2}$  | 43  | -6             | 68  | 2              | 93  | $-\frac{1}{10}$ |
| 19  | $-\frac{3}{2}$ | 44  | $\frac{1}{7}$  | 69  | -2             | 94  | $\frac{2}{9}$   |
| 20  | 4              | 45  | $-\frac{1}{7}$ | 70  | $\frac{7}{2}$  | 95  | $-\frac{2}{9}$  |
| 21  | -4             | 46  | $\frac{1}{3}$  | 71  | $-\frac{7}{2}$ | 96  | $\frac{3}{8}$   |
| 22  | $\frac{1}{5}$  | 47  | $-\frac{1}{3}$ | 72  | 8              | 97  | $-\frac{3}{8}$  |
| 23  | $-\frac{1}{5}$ | 48  | $\frac{3}{5}$  | 73  | -8             | 98  | $\frac{4}{7}$   |
| 24  | $\frac{1}{2}$  | 49  | $-\frac{3}{5}$ | 74  | $\frac{1}{9}$  | 99  | $-\frac{4}{7}$  |
| 25  | $-\frac{1}{2}$ | 50  | 1              | 75  | $-\frac{1}{9}$ | 100 | $\frac{5}{6}$   |

As you can see this function is not one-to-one. The output 1 is attained by the inputs 2 and 10. Despite that this function has no inverse, there is a formula that gives the least input  $n$  when fed the rational number  $p/q$ .

## 2 Smarandache Sequences

The following sequences are called Smarandache crescendo and decrescendo sequences:

Smarandache crescendo sequence: 1, 1, 2, 1, 2, 3, 1, 2, 3, 4, 1, 2, 3, 4, 5, ...

Smarandache decrescendo sequence: 1, 2, 1, 3, 2, 1, 4, 3, 2, 1, 5, 4, 3, 2, 1, ...

The online encyclopedia of integer sequences (also known as the OEIS which can be found at [oeis.org](http://oeis.org)) lists these two sequences. The OEIS designations of these sequences are A002260 and A004736, respectively [1, 2]. In 2003, I found the following formulas for these sequences, in which  $\langle x \rangle := \lfloor x + 1/2 \rfloor$  is the round function:

$$A002260(n) = \frac{2n + \langle \sqrt{2n} \rangle - \langle \sqrt{2n} \rangle^2}{2} \quad (1)$$

$$A004736(n) = \frac{2 - 2n + \langle \sqrt{2n} \rangle + \langle \sqrt{2n} \rangle^2}{2} \quad (2)$$

Note that formulas (1) and (2) are defined at  $n = 0$  as well: since  $\langle \sqrt{0} \rangle = 0$ , they give  $A002260(0) = 0$  and  $A004736(0) = 1$ . The main result of this report is the following formula for  $r(n)$ , defined by a single expression for every  $n \in \mathbb{N}$ , which exhibits a surjection of  $\mathbb{N}$  onto  $\mathbb{Q}$ :

$$r(n) = (-1)^n \cdot \frac{A002260(\lfloor n/2 \rfloor)}{A004736(\lfloor n/2 \rfloor)}. \quad (3)$$

In particular  $r(1) = -0/1 = 0$ , with no special case needed: zero is attained exactly once, at  $n = 1$ . This function's first hundred outputs were given in the introduction. As a pseudoinverse recovering the least  $n$  from a rational number  $p/q$  in lowest terms, we have:

$$s(p/q) = \begin{cases} 2 - p + (p^2 - 3q + 2pq + q^2) & \text{if } p \geq 1 \text{ and } q \geq 1 \\ 3 + p + (p^2 - 3q - 2pq + q^2) & \text{if } p \leq 0 \text{ and } q \geq 1. \end{cases} \quad (4)$$

Note that the second branch covers  $p = 0$ : it gives  $s(0/1) = 3 + 0 + (0 - 3 - 0 + 1) = 1$ , and indeed  $r(1) = 0$ .

### 3 Resolution of the Conjecture

Earlier versions of this document conjectured that  $r(s(p/q)) = p/q$ . The conjecture was resolved by the automated theorem prover Claude Fable 5 [3], which produced the proofs below. We now prove this. First we record the pairing structure underlying (1) and (2).

**Lemma 1.** *For  $m \geq 1$  let  $t$  be the unique positive integer with  $\frac{t(t-1)}{2} < m \leq \frac{t(t+1)}{2}$  (the “row” of  $m$ ), and set  $a = m - \frac{t(t-1)}{2}$ ,  $b = t - a + 1$ . Then:*

- (i)  $t = \langle \sqrt{2m} \rangle$ , and consequently  $A002260(m) = a$  and  $A004736(m) = b$  as given by (1) and (2);
- (ii) the map  $m \mapsto (a, b)$  is a bijection from  $\{1, 2, 3, \dots\}$  onto  $\mathbb{N} \times \mathbb{N}$ , with inverse

$$m = \frac{(a+b-1)(a+b-2)}{2} + a. \quad (5)$$

*Proof.* (i) The defining inequalities are  $t^2 - t < 2m \leq t^2 + t$ . Since  $2m$  is an integer, the left inequality gives  $2m \geq t^2 - t + 1 > (t - \frac{1}{2})^2$ , and the right gives  $2m \leq t^2 + t < (t + \frac{1}{2})^2$ . Hence  $t - \frac{1}{2} < \sqrt{2m} < t + \frac{1}{2}$ , i.e.  $t = \langle \sqrt{2m} \rangle$ . Substituting  $a = m - \frac{t(t-1)}{2} = \frac{2m+t-t^2}{2}$  and  $b = t + 1 - a = \frac{2-2m+t+t^2}{2}$  yields exactly (1) and (2), and these agree with the OEIS definitions of A002260 and A004736, which list the  $t$ -th rows  $1, 2, \dots, t$  and  $t, \dots, 2, 1$  respectively.

(ii) Within row  $t$ ,  $a$  ranges over  $1, \dots, t$  and  $b = t - a + 1$  ranges over  $t, \dots, 1$ , so  $a + b - 1 = t$ . Given any pair  $(a, b)$  of positive integers, it occurs exactly once, namely in row  $t = a + b - 1$  at position  $a$ , which is index (5).  $\square$

**Theorem 1** (Surjectivity). *The function  $r$  defined in (3) maps  $\mathbb{N}$  onto  $\mathbb{Q}$ .*

*Proof.* Let  $x \in \mathbb{Q}$ . If  $x = 0$  then  $x = r(1)$ , since (1) and (2) give  $r(1) = -0/1 = 0$ . Otherwise write  $|x| = p/q$  in lowest terms with  $p, q \geq 1$ . By Lemma 1(ii) there is an  $m \geq 1$  with  $A002260(m) = p$  and  $A004736(m) = q$ . Then  $r(2m) = p/q$  and  $r(2m + 1) = -p/q$ , so  $x$  is attained either way.  $\square$

**Theorem 2** (The conjecture). *Let  $p/q$  be in lowest terms with  $q \geq 1$ . Then  $s(p/q) \in \mathbb{N}$ ,*

$$r(s(p/q)) = \frac{p}{q}, \quad (6)$$

*and  $s(p/q)$  is the least  $n \in \mathbb{N}$  with  $r(n) = p/q$ .*

*Proof.* If  $p = 0$  then  $s(0/1) = 1$  by the second branch of (4), and  $r(1) = 0$ . Moreover  $r(n) = 0$  forces  $A002260(\lfloor n/2 \rfloor) = 0$ , which by Lemma 1 happens only for  $\lfloor n/2 \rfloor = 0$ , i.e.  $n = 1$ ; so 1 is the unique (hence least) preimage of 0. Now assume  $p \neq 0$  and put  $P = |p|$ . By Lemma 1(ii) the pair  $(P, q)$  occurs at index

$$m_0 = \frac{(P+q-1)(P+q-2)}{2} + P. \quad (7)$$

Expanding with  $\sigma = P + q$ :

$$2m_0 = \sigma^2 - 3\sigma + 2 + 2P = P^2 + 2Pq + q^2 - 3q - P + 2. \quad (8)$$

If  $p \geq 1$  (so  $P = p$ ), this says  $2m_0 = 2 - p + (p^2 - 3q + 2pq + q^2) = s(p/q)$ , and  $r(2m_0) = (+1) \cdot P/q = p/q$ . If  $p \leq -1$  (so  $P = -p$ ), then  $2m_0 + 1 = 3 + p + (p^2 - 3q - 2pq + q^2) = s(p/q)$ , and  $r(2m_0 + 1) = (-1) \cdot P/q = p/q$ . In both cases  $s(p/q)$  is a natural number of the correct parity and  $r(s(p/q)) = p/q$ .

For minimality, suppose  $r(n) = p/q$ . Since  $p \neq 0$  while  $r(1) = 0$ , necessarily  $n \geq 2$ . Let  $m = \lfloor n/2 \rfloor$ ,  $(a, b) = (A002260(m), A004736(m))$ . Then  $a/b = P/q$  with  $p/q$  reduced forces  $(a, b) = (kP, kq)$  for some  $k \geq 1$ . By (5), the index of  $(kP, kq)$  is strictly increasing in  $k$  (its row  $k(P+q) - 1$  grows with  $k$ ), so  $m \geq m_0$ , with equality iff  $k = 1$ . Since the sign of  $r(n)$  fixes the parity of  $n$ , the least admissible  $n$  is  $2m_0$  when  $p > 0$  and  $2m_0 + 1$  when  $p < 0$  — precisely  $s(p/q)$ .  $\square$

## 4 Conclusion

The map  $r$  is a surjection of  $\mathbb{N}$  onto  $\mathbb{Q}$  (Theorem 1), and the pseudoinverse  $s$  satisfies  $r(s(p/q)) = p/q$  and returns the least preimage (Theorem 2). The underlying mechanism is the classical diagonal (Cantor-style) pairing: A002260 and A004736 are the two coordinates of the inverse pairing function, and the closed form for  $s$  is the pairing polynomial (5) in disguise. Since  $r$  is a surjection and  $\mathbb{N} \hookrightarrow \mathbb{Q}$ , it follows (without choice, by the explicit  $s$ ) that  $\mathbb{N}$  and  $\mathbb{Q}$  are equipollent.

## Acknowledgments

The author posed the conjecture  $r(s(p/q)) = p/q$  and supplied the formulas (1), (2), (3), and (4). The proof of the conjecture, comprising Lemma 1 and Theorems 1–2, was discovered by Claude Fable 5 (Anthropic) [3] in July 2026, prompted with the conjecture and the formulas above. The author checked the proof by hand and verified the identities computationally.

## References

- [1] OEIS Foundation Inc. (2003), Entry A002260 in *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A002260>
- [2] OEIS Foundation Inc. (2003), Entry A004736 in *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A004736>

- [3] Anthropic (2026). *Claude Fable 5* [Large language model]. Used as an automated theorem prover to resolve the conjecture of Section 3. <https://www.anthropic.com/news/claude-fable-5-mythos-5>